EE4-26
EE9-CS5-3
EE9-FPN2-07

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2018

MSc and EEE PART IV: MEng and ACGI

**Corrected copy**

# ESTIMATION AND FAULT DETECTION

Wednesday, 9 May 10:00 am

Time allowed: 3:00 hours

**There are FOUR questions on this paper.**

**Answer ALL questions.**

*All questions carry equal marks*

**Any special instructions for invigilators and information for candidates are on page 1.**

Examiners responsible First Marker(s) : T. Parisini
Second Marker(s) : D. Angeli

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# ESTIMATION AND FAULT DETECTION

Information for candidates:

- **One-step ahead Kalman predictor:**

$$\hat{x}(t+1|t) = F\hat{x}(t|t-1) + K(t)[y(t) - H\hat{x}(t|t-1)]$$

- **Kalman predictor gain**

$$K(t) = FP(t)H^{\top}\left(V_2 + HP(t)H^{\top}\right)^{-1}, \quad t = 1,2,\ldots$$

- **Riccati equation**

$$P(t+1) = F\left[P(t) - P(t)H^{\top}\left(V_2 + HP(t)H^{\top}\right)^{-1}HP(t)\right]F^{\top} + V_1, \quad t = 1,2,\ldots$$

- **Realization: observer canonical form**

  Given

$$Y(s)/U(s) = \frac{b_m s^m + \cdots + b_1 s + b_0}{s^n + a_{n-1}s^{n-1} + \cdots a_1 s + a_0} \quad \text{with} \quad m < n$$

  then:

$$\begin{cases} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 1 & 0 & -a_{n-2} \\ 0 & \cdots & 0 & 1 & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_m \\ 0 \\ \vdots \\ 0 \end{bmatrix} u \\ y = [0 \cdots 0\,1] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} \end{cases}$$

- **Generic Structure of an Observer**

$$\begin{cases} \dot{z} = Fz + TBu + Ky \\ \hat{x} = z + Hy \end{cases}$$

- **Basic Conditions for UIOs**

$$(I - HC)N = 0\,, \quad N: \text{disturbance distribution matrix;}$$
$$F = A - HCA - K_1C;\ K = K_1 + FH;\ T = I - HC$$

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1. Consider the mechanical system drawn in Fig. 1.1.

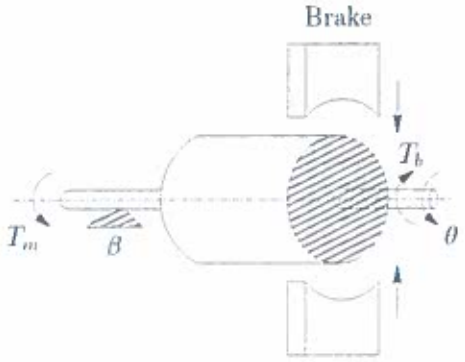



Figure 1.1 Scheme of the mechanical system of Question 1.

The mechanical shaft is connected to the cylinder with a known total inertia moment $J$ and the rotation is imposed by the known motor torque $T_m$. The angular position is denoted by $\theta$ and is perfectly measurable by an ideal encoder. The linear viscous friction torque is characterised by a known friction parameter $\beta$. At time $t = t_0$, the braking system is activated generating a *constant and unknown* braking torque $T_b$.

a) Determine a state-space representation of the system valid for $t < t_0$ (that is, before the action of the braking system) supposing that the input is given by $u(t) = T_m(t)$ and the output is given by $y(t) = \theta(t)$.

**[ 4 marks ]**

b) For all $t < t_0$, discuss the observability properties from the output $y(t)$.

**[ 3 marks ]**

c) Consider the system for $t \geq t_0$ after the action of the braking system, that is consider the effect of the unknown and constant braking torque $T_b$. Establish whether it is possible to devise an observer scheme to estimate *simultaneously* the braking torque $T_b$ and the state $x(t)$ associated with the state-space description derived in your answer to Question 1-a).

*Hint: do not attempt to design such an observer.* **[ 6 marks ]**

d) In case the observer design of Question 1-c) is possible, denote by $\hat{x}(t)$ the estimate of the state $x(t)$ associated with the state-space description derived in your answer to Question 1-a) and denote by $\hat{T}_b(t)$ the estimate of the braking torque $T_b$ described in Question 1-c). Let $e(t) := [(x(t) - \hat{x}(t))^\top, T_b(t) - \hat{T}_b(t)]^\top$ denote the total estimation error; moreover, suppose that its dynamics obeys

$$\dot{e}(t) = Fe(t).$$

Denote by $\lambda_i, i = 1, \ldots, n$ the eigenvalues of $F$, where $n$ is the order of the observer scheme devised in your answer to Question 1-c). Suppose that the values of the mechanical parameters are given by: $J = 10$, $\beta = 1$. Design an $n$-order state observer such that

$$\lambda_i = -1, i = 1, \ldots, n.$$

**[ 7 marks ]**

2. Consider the discrete-time dynamic system described in Fig. 2.1

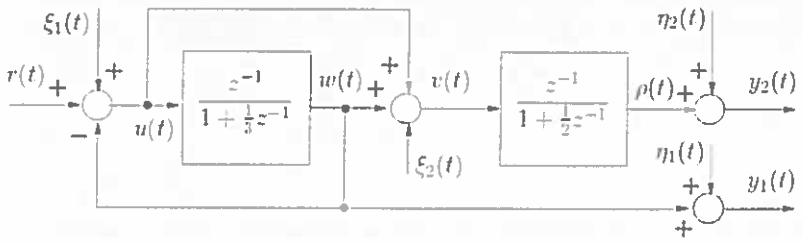



Figure 2.1 Block diagram for Question 2.

where $r(t)$ is a deterministic input variable, $\xi_1(\cdot) \sim WGN(0,1)$, $\xi_2(\cdot) \sim WGN(0,1)$, $\eta_1(\cdot) \sim WGN(0,9)$, $\eta_2(\cdot) \sim WGN(0,4)$ (Gaussian zero-mean stochastic processes) and the stochastic processes $\xi_1(\cdot)$, $\xi_2(\cdot)$, $\eta_1(\cdot)$, $\eta_2(\cdot)$ are supposed to be independent of each other.

a) Referring to the system sketched in Fig. 2.1, determine a stochastic state-space representation of the system supposing that the deterministic input is given by $r(t)$ and the output by $y(t) := [y_1(t), y_2(t)]^\top \in \Re^2$, respectively.

**[ 4 Marks ]**

b) Consider the one-step ahead optimal steady-state Kalman predictor of the state $x$ for the state-space description in Question 2-a). Using the asymptotic theory of steady-state Kalman estimation, show that the Algebraic Riccati Equation admits a feasible matrix solution $\bar{P}$ and compute it. Compute the corresponding gain vector $\bar{K}$ and write the difference equation yielding the steady-state Kalman prediction $\hat{x}(t|t-1)$.

**[ 5 Marks ]**

c) Compute the covariance matrix of the prediction error, $\mathrm{Cov}[x(t) - \hat{x}(t|t-1)]$, and the covariance matrix of the process, $\mathrm{Cov}[x(t)]$. Compare $\mathrm{Cov}[x(t) - \hat{x}(t|t-1)]$ with $\mathrm{Cov}[x(t)]$ and comment on your findings.

**[ 5 Marks ]**

d) Consider the optimal steady-state Kalman filter yielding $\hat{x}(t|t)$. From your answer to Question 2-b), compute the constant gain vector $\bar{K}_0$ of the optimal steady-state Kalman filter. Compute the covariance matrix of the filtering error, $\mathrm{Cov}[x(t) - \hat{x}(t|t)]$ and compare it with $\mathrm{Cov}[x(t) - \hat{x}(t|t-1)]$ and $\mathrm{Cov}[x(t)]$ computed in your answer to Question 2-c). Comment on your findings.

**[ 6 Marks ]**

3. Consider the continuous-time dynamic system depicted in Fig. 3.1.

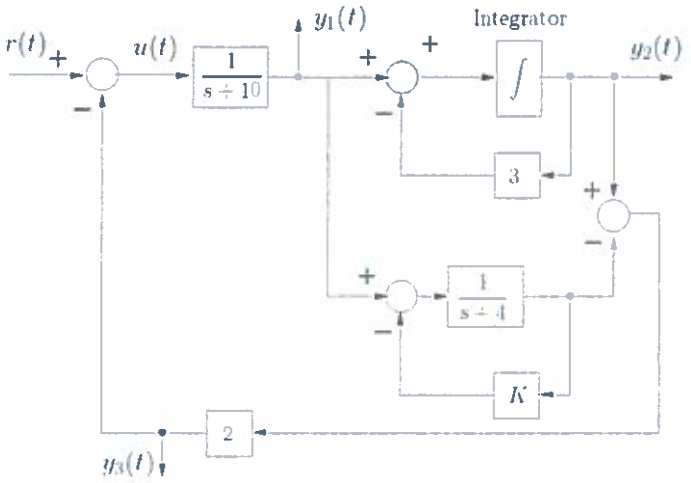



Figure 3.1 Block diagram for Question 3.

The system depicted in Fig. 3.1 has one input variable $r(t)$, three output variables $y_1(t)$, $y_2(t)$, and $y_3(t)$ and it also comprises transfer functions, an integrator, and several "gain" blocks (their output is given their input multiplied by the scalar shown inside the block). Moreover, $K \in \Re$ is a parameter.

a) Determine the transfer functions $G_1(s)$, $G_2(s)$, and $G_3(s)$ from the input $r(t)$ and the outputs $y_1(t)$, $y_2(t)$, and $y_3(t)$, respectively.

**[ 5 marks ]**

b) Determine a state-space representation of the system in Fig. 3.1 supposing that the input variable is $r(t)$ and the output is given by $y(t) := [y_1(t), y_2(t), y_3(t)]^\top \in \Re^3$.

**[ 3 marks ]**

c) Analyse the observability of the whole system in Fig. 3.1 from each output $y_i(t)$, $i = 1, 2, 3$ *taken separately* as a function of the parameter $K \in \Re$. Comment on your findings taking also into account your answer to Question 3-a).

**[ 5 marks ]**

d) Consider the value of $K$ determined in your answer to Question 3-c) making the system in Fig. 3.1 not observable from one of the outputs. Select this output variable and, *disregarding the other output variables*, for the above value of $K$ determine a state-space description which is equivalent to that determined in your answer to Question 3-b) and in which the observable and the non-observable sub-systems are clearly identified.

**[ 7 marks ]**

4. Consider the continuous-time dynamic system described by the following state equations:

$$\begin{cases} \dot{x}_1 = -3x_1 + x_2 + u + d \\ \dot{x}_2 = x_1 - 4x_2 + f \\ y_1 = x_1 + x_2 \\ y_2 = 2x_1 + x_2 \end{cases} \tag{4.1}$$

where $x = [x_1, x_2]^\top$ is the state vector, $y = [y_1, y_2]^\top$ is the output vector, $u$ is a *known* input, $d$ is an *unknown* disturbance, and $f$ denotes a fault affecting the second state equation for $t \geq T_0$, where $T_0$ denotes the *unknown* time of fault occurrence. Suppose that $f(t) = 0, \forall t < T_0$ and assume that only one fault may occur during the whole time-horizon $t \in (0, \infty)$.

a) Consider the system (4.1) and suppose that no disturbance $d(t)$ nor fault $f(t)$ act on the system at any time, that is, suppose that $d(t) = 0, \forall t$ and $f(t) = 0, \forall t$. Denoting by $\widehat{x}(t)$ an estimate of the state $x(t)$ of the system (4.1), let $e(t) = x(t) - \widehat{x}(t)$ denote the state estimation error and suppose that its dynamics obeys

$$\dot{e}(t) = Fe(t),$$

where $\lambda_1, \lambda_2$ are the eigenvalues of $F$. Design a full-order state observer such that $\lambda_1 = -1, \lambda_2 = -2$. *Hint: the design of the observer matrix gain admits more than one solution. It is suggested to suitably restrict the number of free parameters of the observer gain.* **[ 5 marks ]**

b) Suppose that no fault $f(t)$ acts on the system at any time, that is, suppose that $f(t) = 0, \forall t$. Using the state observer designed in your answer to Question 4-a), consider the time behaviour of the output residual

$$\varepsilon(t) = Ce(t), \ \forall t \geq 0$$

for a given value $\bar{e}_0$ of the initial estimation error $e(0)$ and assume that the disturbance is given by

$$d(t) = 0.1 \sin(t) \cdot 1(t),$$

where $1(t)$ denotes the unit step function. Determine the expression of the Laplace transform $\mathscr{L}[\varepsilon(t)]$. **[ 5 marks ]**

c) Consider the system (4.1) for any $t \geq 0$ under the action of the disturbance considered in Question 4-b), that is, $d(t) = 0.1 \sin(t) \cdot 1(t)$ and also suppose that a fault $f$ occurs at time $T_0 = 20$sec taking on the following form:

$$f(t) = 2 \cdot 1(t - 20).$$

Design an *Unknown Input Observer* (UIO) such that the output residual $\widetilde{\varepsilon}(t)$ is insensitive to the disturbance $d(t)$ but instead is sensitive to the fault $f$ and where the eigenvalues characterising the dynamics of the estimation error are the same as in Question 4-a), that is $\lambda_1 = -1, \lambda_2 = -2$. *Hint: Similarly to Question 4-a), the design of the observer matrix gain admits more than one solution. It is suggested to suitably restrict the number of free parameters of the observer gain.* **[ 6 marks ]**

d) Determine the expression of the Laplace transform $\mathscr{L}[\widetilde{\varepsilon}(t)]$ when the UIO is used and compare it with the expression of the Laplace transform $\mathscr{L}[\varepsilon(t)]$ determined in your answer to Question 4-b). From this comparison, characterise the qualitative time-behaviour of $\varepsilon(t)$ and $\widetilde{\varepsilon}(t)$ before the occurrence of the fault, that is, for $t < 20$, and comment on your findings in terms of the design of a fault detection scheme using the observers designed in your answers to Question 4-b) and Question 4-c), respectively. **[ 4 marks ]**